

Quiz # 3

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Subject: Applied Physics

Zener diode:

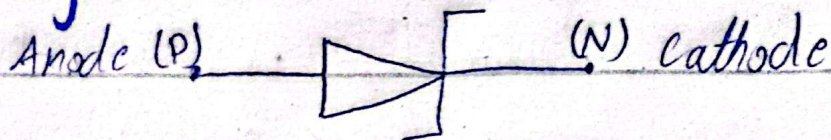
A zener diode is made from is a special type of semiconductor diode that is mainly designed to work in reverse bias condition. It is widely used for voltage regulation, voltage stabilization, and protection circuit.

Why it is heavily doped:-

Zener diodes are heavily doped on both the P-side and N-side to create a very thin depletion layer. A thin depletion layer means a very intense electric field can develop across it even at relatively low reverse voltages, enabling breakdown (zener effect) to occur at a precisely controlled, low voltage. An ordinary PN diode is lightly doped,

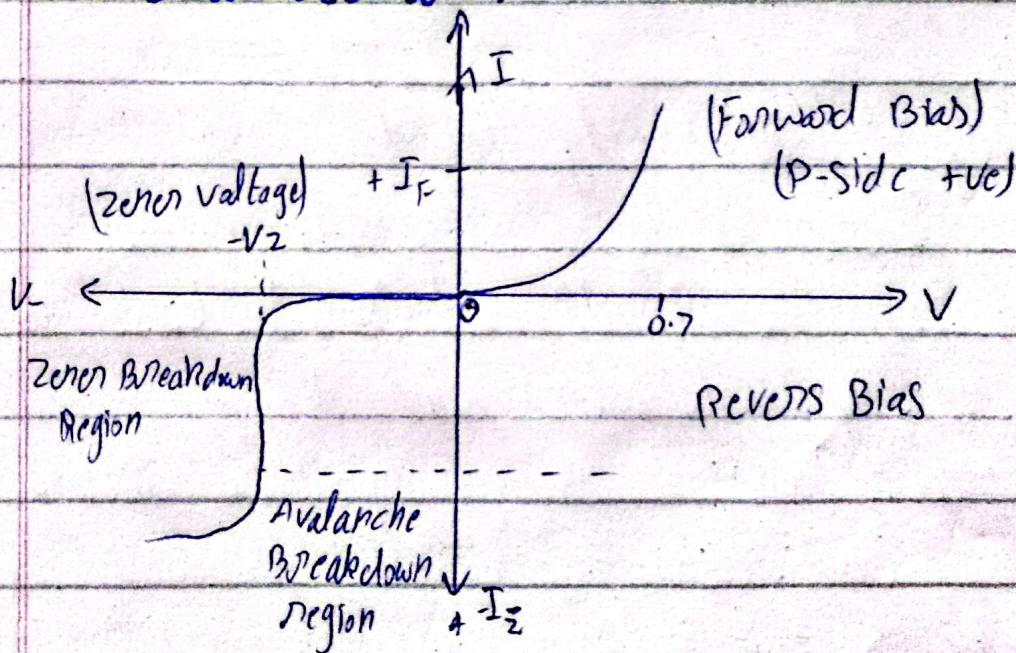
resulting in a wider depletion region and a much higher (and less controlled) break down voltage.

Symbol:



(cathode line is bend from the edges)

V-I Characteristic:



Explain V I Characteristic:

Forward Bias region (1st Quadrant)
The diode behaves like a normal

silicon diode. After about 0.7V (Si) current increase rapidly.

Reverse Bias (III Quadrant)

Initially, only a small leakage current (I_s) flows.

At $V = -V_Z$ (Zener voltage)

The diode enters breakdown region.

A large reverse current flows while the voltage remains almost constant at $-V_Z$.

Zener breakdown (low V_Z)

The breakdown occur at lower voltage and the curve is very sharp.

In Avalanche breakdown (high V_Z)

Breakdown occurs at higher voltage.

Safe Operation: The current must be limited by an external resistor so that diode does not get damage.

Why Voltage remain constant in Break down?

In the breakdown region, two quantum-mechanical mechanisms work together to

keep the voltage across zener diode nearly constant.

Zener effect: ($V_Z < 5V$)

The very thin depletion layer (due to heavy doping) allows electrons to quantum-mechanically tunnel directly from the valence band of the P-side to the conduction band of the N-side without needing activation energy. This generates a large reverse current at a well defined voltage.

Avalanche Effect: ($V_Z > 5V$)

carriers gain enough energy to ionise atoms, liberating more electrons in a chain reaction, which increases current while keeping voltage steady.

Once in breakdown, the Zener diode has very low dynamic resistance, so large changes in current produce only tiny changes in voltage. Each Excess voltage drops across an external series resistor, keeping V_Z constant - making it ideal for voltage regulation.